

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Analytics Project — Python, SQL & Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows:** 3,900
- Columns:** 18
- Key Features:**
 - Customer demographics — Age, Gender, Location, Subscription Status
 - Purchase details — Item Purchased, Category, Purchase Amount, Season, Size, Color
 - Shopping behavior — Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- Missing Data:** 37 values in the Review Rating column

3. Exploratory Data Analysis using Python

Data preparation and cleaning were carried out in Python before any analysis began:

- Data Loading:** Imported the dataset using `pandas`.
- Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

Output of `df.describe(include='all')` — Part 1 of 2

Stat	Customer ID	Age	Gender	Item Purch.	Category	Purch. Amt	Location	Size	Color	Season	Review Rtg	Subscr.	Shipping
count	3900.00	3900.00	3900	3900	3900	3900.00	3900	3900	3900	3900	3863.00	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Ship.
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675
mean	1950.50	44.07	NaN	NaN	NaN	59.76	NaN	NaN	NaN	NaN	3.75	NaN	NaN
std	1125.98	15.21	NaN	NaN	NaN	23.69	NaN	NaN	NaN	NaN	0.72	NaN	NaN
min	1.00	18.00	NaN	NaN	NaN	20.00	NaN	NaN	NaN	NaN	2.50	NaN	NaN
25%	975.75	31.00	NaN	NaN	NaN	39.00	NaN	NaN	NaN	NaN	3.10	NaN	NaN
50%	1950.50	44.00	NaN	NaN	NaN	60.00	NaN	NaN	NaN	NaN	3.80	NaN	NaN
75%	2925.25	57.00	NaN	NaN	NaN	81.00	NaN	NaN	NaN	NaN	4.40	NaN	NaN
max	3900.00	70.00	NaN	NaN	NaN	100.00	NaN	NaN	NaN	NaN	5.00	NaN	NaN

Output of `df.describe(include='all')` — Part 2 of 2

Stat	Discount Appl.	Promo Code Used	Prev. Purchases	Payment Method	Freq. of Purch.
count	3900	3900	3900.00	3900	3900
unique	2	2	NaN	6	7
top	No	No	NaN	PayPal	Every 3 Months
freq	2223	2223	NaN	677	584
mean	NaN	NaN	25.35	NaN	NaN
std	NaN	NaN	14.45	NaN	NaN
min	NaN	NaN	1.00	NaN	NaN
25%	NaN	NaN	13.00	NaN	NaN
50%	NaN	NaN	25.00	NaN	NaN
75%	NaN	NaN	38.00	NaN	NaN

Stat	Discount Appl.	Promo Code Used	Prev. Purchases	Payment Method	Freq. of Purch.
max	NaN	NaN	50.00	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to `snake_case` for better readability and documentation.
- **Feature Engineering:**
 - Created `age_group` column by binning customer ages.
 - Created `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified whether `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in PostgreSQL to answer ten key business questions:

1. Revenue by Gender — Compared total revenue generated by male vs. female customers.

Gender	Revenue (USD)
Female	75,191
Male	157,890

2. High-Spending Discount Users — Identified customers who used discounts but still spent above the average purchase amount — 839 customers matched.

Customer ID	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97

(Showing top 5 of 839 matching rows.)

3. Top 5 Products by Rating — Found the products with the highest average review ratings.

Item Purchased	Avg. Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison — Compared average purchase amounts between Standard and Express shipping.

Shipping Type	Avg. Purchase Amount (USD)
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers — Compared average spend and total revenue across subscription status.

Subscription Status	Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645.00
No	2,847	59.87	170,436.00

6. Discount-Dependent Products — Identified the 5 products with the highest percentage of discounted purchases.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation — Classified customers into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3,116
New	83
Returning	701

8. Top 3 Products per Category — Listed the most purchased products within each category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions — Checked whether customers with more than 5 purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2,518
Yes	958

10. Revenue by Age Group — Calculated the total revenue contribution of each age group.

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Dashboard in Power BI

An interactive PowerBI dashboard was built to present the insights visually, with filters for subscription status, gender, category, and shipping type.



6. Business Recommendations

- **BoostSubscriptions:** Promote exclusive benefits for subscribers to grow the current 27% subscriber base.
- **Customer Loyalty Programs:** Reward repeat buyers to move them from the "Returning" into the "Loyal" segment.
- **Review Discount Policy:** Balance sales boosts from heavily discounted items (e.g., Hat, Sneakers, Coat) against margin control.
- **Product Positioning:** Highlight top-rated and best-selling products — such as Gloves, Jewelry, and Blouse — in marketing campaigns.
- **Targeted Marketing:** Focus efforts on high-revenue age groups (Young Adult, Middle-aged) and Express-shipping users.

Prepared using Python (pandas), PostgreSQL, and Power BI — full scripts and dashboard files available in the project's GitHub repository.